Topic: Selection Operation

Q. Explain the Selection operation in Relational Algebra. Write its syntax and provide examples demonstrating how it filters tuples.

> 📌 **Definition to Remember**
> The **Selection Operation** (denoted by σ) is a **unary operation** in relational algebra that filters and returns specific **rows (tuples)** from a relation that satisfy a given logical condition (predicate). It is equivalent to the `WHERE` clause in SQL.

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1. Selection Operator (σ — Sigma)

- Operates on a **single table** (unary).
- Filters **horizontally** — returns same columns, but only matching rows.
- Output automatically eliminates duplicate rows (set behavior).

2. Syntax of Selection

$σ_{}(Relation_Name)$

- **Predicate:** Uses comparison operators (=, ≠, <, ≤, >, ≥) and logical operators (AND AND, OR OR, NOT NOT).

3. Examples

Input Relation: EMPLOYEE

Emp_ID	Name	Department	Salary
1	Alice	IT	50000
2	Bob	HR	40000
3	Charlie	IT	60000
4	David	Sales	45000

Example 1: Simple Condition

Query: Find all IT department employees.
$σ_{Department = 'IT'}(EMPLOYEE)$
Result: Returns rows for Alice (ID 1) and Charlie (ID 3).

Example 2: AND Condition (AND)

Query: IT employees with salary > 55000.
$σ_{Department = 'IT' AND Salary > 55000}(EMPLOYEE)$
Result: Returns only Charlie (IT, 60000).

Example 3: OR Condition (OR)

Query: Employees in HR or Sales.
$σ_{Department = 'HR' OR Department = 'Sales'}(EMPLOYEE)$

Result: Returns Bob (HR) and David (Sales).

⭐ **Must-Write Points (for 10 marks)**

1. Selection is a **unary operation** — operates on one relation.
2. Denoted by **σ (Sigma)**; equivalent to SQL's `WHERE` clause.
3. Filters **rows (tuples)** based on a logical predicate — horizontal filtering.
4. Output has the same columns as input but only matching rows.
5. Predicates use comparison operators (=, ≠, <, >) and logical operators (∧, ∨, ¬).
6. Can be **nested**: apply selection to the result of another operation.
7. Automatically eliminates duplicate rows (set semantics).

⚡ **Quick Recall**

`σ (Sigma) → Unary → Horizontal Filtering (rows) → Predicate (=, ≠, ∧, ∨) → Equivalent to SQL WHERE → Can be nested`